Aryan Isaac Bhobe

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EDUCATION

University of Bath Oct 2023 – Jul 2027
B.Sc (Hons) Computer Science & Artificial Intelligence Bath, United Kingdom

- Current Classification – **74.50% | First Class Honours**
- Relevant Courses – Corporate Finance, Risk Management, Business Strategy, Probability, Statistics, Linear Algebra, Databases, Natural Language Processing, Bayesian Machine Learning, Reinforcement Learning, Algorithms & Data Structures, Software Engineering

EXPERIENCE

Accenture Jun 2026 – Sep 2026
AI & Data Summer Analyst London, United Kingdom

- Led design of C-suite thought leadership on digital transformation and AI for the Aerospace & Defence practice, **recognised by the Global Lead for Maritime** (presented to 35+ global leadership members) and slated for reuse in conferences and marketing material.
- Synthesised 15+ RFI responses, credentials and leadership articles into a commercial view of the industry's digital and AI priorities, and prepared annotated diagrams and visuals of technical implementations.
- Engineered a multi-agent AI codebase documentation system (LangGraph) used by 4 engineers, **accelerating project by ∼1.5 weeks**.
- Orchestrated specialist agents in sequential and parallel workflows, with MCP tool servers and an async FastAPI layer (Supabase caching) exposing data sources, and Pydantic schemas plus prompt engineering resulting in **0 schema failures in 50 test runs**.

Merck Sharp & Dohme (MSD) Jun 2025 – Jun 2026
Machine Learning Scientist London, United Kingdom

- Built a modular, documented statistical QC pipeline (PCA and z-score outlier detection, kNN/Random Forest imputation) for high-dimensional data, **adopted as departmental standard** and used as the **basis for biomedical analysis AI agent training**.
- Evaluated 20+ bioinformatics and ML research papers to shortlist methods, then selected the approach that optimised ARI, BIC and silhouette scores, **improving cluster stability** (validated by bootstrap resampling and cross-checked against HDBSCAN).
- Led independent ML research on 550k samples across 18 relational tables, using K-Means, GMM and HDBSCAN with MCA dimensionality reduction on GPU-accelerated HPC to uncover clinically meaningful patient subtypes.
- Communicated proteomics results to Neuroscience senior leadership and wet-lab biologists through slide decks, spider plots and heatmaps, prompting follow-up requests. **Presented methodology to VP in Epidemiology**.
- **Credited as co-author** on a scientific poster at a neuroscience symposium in Boston for data analysis and visualisation, including time-series modelling across cohorts (regression, hierarchical clustering) and hypothesis testing.

Spookfish Innovations Jul 2024 – Sep 2024
Deep Learning Engineer Intern Bengaluru, India

- Developed a video ad-insertion system in an agile ML team, building the final pipeline (PyTorch, OpenCV) that replaces content across frames using segmentation masks, with temporal consistency through object tracking.
- Trained and optimised segmentation models (YOLOv9, RTMDet, Mask R-CNN) via transfer learning and augmentation, improving accuracy across IoU 0.5 to 0.95; tracked experiments in Weights & Biases and **handed over documented PoC to engineers**.

VOLUNTEERING & LEADERSHIP

Consultant | *180 Degrees Consulting Bath* Oct 2026 – Present

- Supporting a **time-critical digital migration** for a youth charity client as part of the digital transfer workstream: evaluating replacement platforms and planning the transfer of its web presence against a fixed deadline.
- Designing an impact reporting framework with defined KPIs and tailored reporting templates, and recommending governance processes so management, funders and other stakeholders receive consistent data.

Undergraduate Peer Mentor | *University of Bath, Department of Computer Science* Sep 2024 – Jul 2025

- Mentoring 10 first-year computer science students through regular check-ins, proactive contact ahead of key deadlines and signposting to relevant university services where needed. Received **100% mentee satisfaction**
- Created study resources, ran group sessions and set up a peer support channel matching students on similar modules.

PROJECTS & SKILLS

Commodity Market Analytics & Multi-Agent Research | *Code · Demo* Python | statsmodels | arch | FastMCP

- Prompted by the share price chart of a personal holding, tested a perceived annual cycle in a commodities trader's shares and showed it was **not genuine seasonality** (calendar F-test p=0.71), tracing it to dividend timing and reporting the negative result.
- Found that **copper price moves lead the stock's returns** (Granger causality p=0.006) and confirmed a leverage effect in volatility (GJR-GARCH, p=0.008), but no cointegration (p=0.27).
- **Built a multi-agent research platform** (OpenAI Agents SDK, FastMCP), letting non-technical users query the results in natural language, and deployed it live using Streamlit, resolving MCP environment and timeout issues.

Technical Skills – Python, SQL, Excel, PowerBI, Git; PyTorch, OpenCV, statsmodels; LangGraph, MCP, FastAPI; AWS, Azure